

Challenge Problems Week 6

Quantum Error Correction

UCQ Education Team

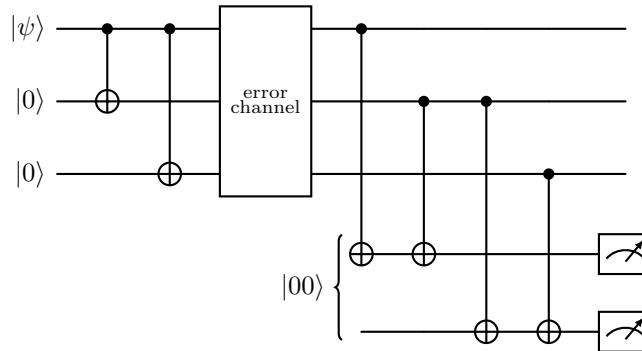
Problem 1 Using the Schrödinger equation, derive the *von Neumann equation*, which describes the time evolution of density matrices

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho]$$

where the bracket is the commutator.

(In more advanced physics, this equation can also be derived by applying canonical quantization to *Liouville's equation* $\frac{\partial \rho}{\partial t} = \{H, \rho\}$. With this approach, we can derive the Schrödinger equation simply by quantizing classical mechanics!)

Problem 2 Consider the *bit flip code*



where the error channel randomly applies X gates to the three qubits. Using the measurement outcomes, show that any single-qubit error can be corrected.